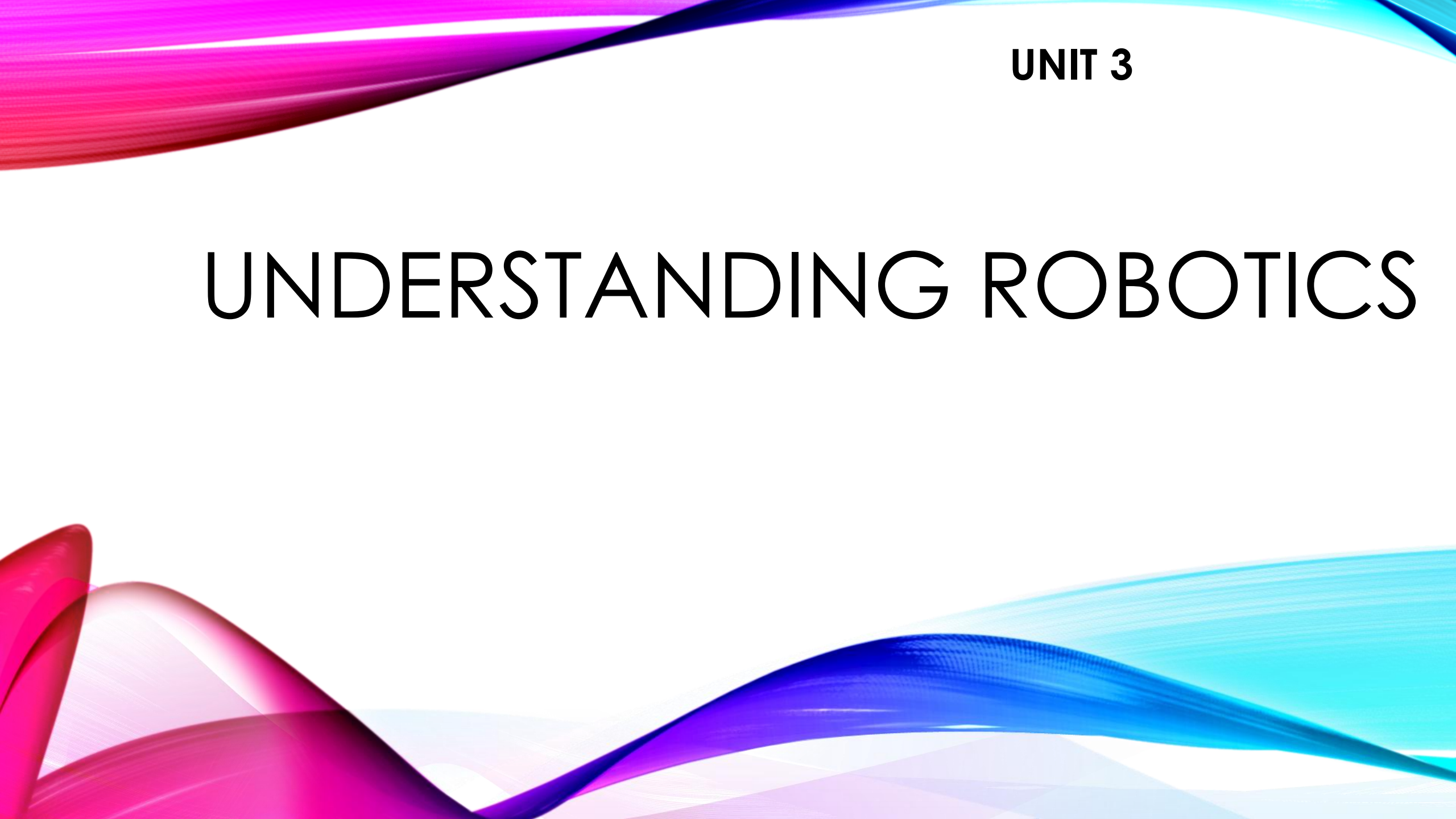




UNIT 3

UNDERSTANDING ROBOTICS

Objectives

- Introduction to Robotics
- History of Robotics
- Types of Robots
- Laws of Robotics

WHAT IS ROBOTICS ?

Robotics is the branch of technology that deals with the design, construction, operation, and use of robots.

A robot is a machine that can carry out tasks autonomously or semi-autonomously, often controlled by computer programs.

Importance of Robotics

- Robotics plays a significant and growing role in various aspects of daily life, impacting society in numerous positive ways.
- Robots assist doctors in performing surgeries more precisely.
- Robots help us learn about technology and programming.
- Robots are used in gaming and entertainment, enhancing virtual and augmented reality experiences.



History of Robotics

- The term "**Robot**" was introduced in 1920 by **Czech** writer Karel Capek in his play "**R.U.R.**" (**Rossum's Universal Robots**).
 - In the 1960s, the first real industrial robot was created, and it was called **Unimate**.
 - **Unimate** was like a super strong and precise arm that could do repetitive tasks in a factory.
 - In the (**1970s-80s**), it was a period where scientists set up special places called robotics research institutes.
 - They focused on things like teaching robots to think and learn.
 - In the (**1990s-2000s**), robots got even smarter.
 - They started using sensors to see and feel their surroundings.
- Plus, they started making decisions on their own, thanks to something called **Artificial Intelligence (AI)**.



Types Of Robots

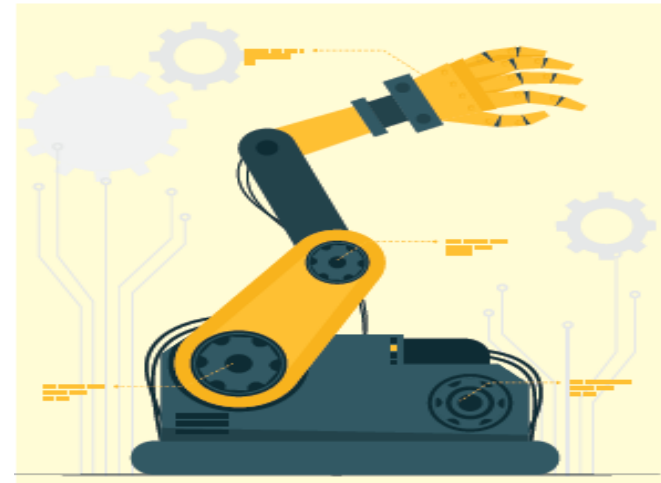
There are various types of robots designed for different purposes and applications.

Industrial Robots

These robots are like super strong, precise arms that help build things in factories.

The arm-like structure of a robot, called the **manipulator**, consists of joints and links. It carries end effectors, such as grippers or welding tools, to perform various tasks.

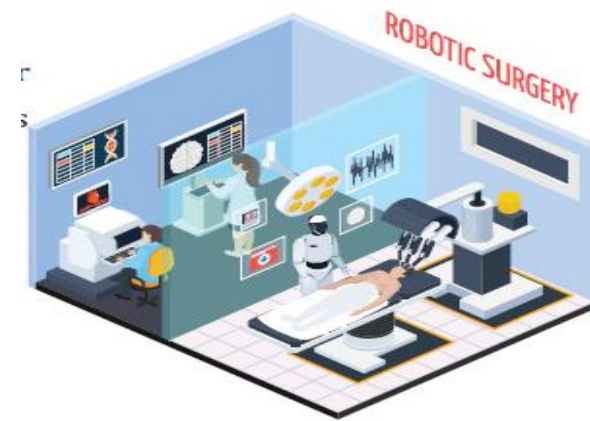
Example: Robotic arms in car manufacturing plants.



- **Medical Robots**

Robots used in medicine can assist surgeons in surgeries or deliver medications.

Robots are more precise and less invasive.



- **Educational Robots**

They can help you understand and practice different subjects, especially in science and programming.



Entertainment Robots

They help in playing games along with you, or even robots used in theme parks.



Domestic Robots

These robots help with tasks at home.

They can vacuum the floors and robotic lawn mowers.



Agricultural Robots

Assist in agricultural tasks such as planting, harvesting, and monitoring crops.

Humanoid Robots

Humanoids are robots that look and act like humans.

They mimic human behavior, and can perform activities, such as running, jumping, carrying objects.

Some of them can mimic human expressions as well.

Robots have sensors embedded and have cameras that enables them to see clearly.

Example: Sophia is a humanoid which exhibits human like expressions and can interact with people.



- **Key Components of Robotics**

Robots consist of various components that work together to perform specific tasks.

Power Source

Robots need a power source to operate. This can be in the form of batteries, electric power, or in some cases, hydraulic or pneumatic systems.

Actuators

Actuators are responsible for the movement of robot parts. Electric motors, hydraulic cylinders, or pneumatic pistons act as actuators, allowing the robot to move its joints or other components.

Sensors

Sensors provide robots with the ability to perceive and gather information about their environment.

Common sensors include:

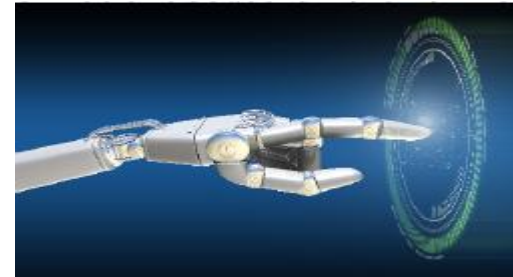
Cameras: For visual information.

Microphones: For sound detection.

Infrared Sensors: For proximity and temperature sensing.

Touch Sensors: To detect physical contact.

Gyroscopes and Accelerometers: For measuring orientation and acceleration.



Control System

The control system is the brain of the robot.

It processes information from sensors, makes decisions, and sends commands to actuators.

It can be a microcontroller or a more sophisticated computer based system.

Feedback Mechanism

A feedback mechanism allows the robot to adjust its actions based on the information received from sensors.

This helps ensure accuracy and precision in performing tasks.

THREE LAWS OF ROBOTICS

The Three Laws of Robotics were introduced by **Isaac Asimov**.

Asimov developed these laws as a set of ethical guidelines for robots, outlining principles to ensure the responsible use of artificial intelligence. The laws are as follows

First Law

A robot may not injure a human being, or through inaction, allow a human being to come to harm.

This law emphasizes the paramount importance of human safety. It dictates that a robot's primary duty is to ensure the well-being of humans and prevent harm to them.

Second Law

A robot must obey orders given to it by human beings, except where such orders would conflict with the First Law.

The second law is the concept of obedience to human commands.

Robots are expected to follow instructions from humans as long as those instructions do not lead to harm or conflict with the First Law.



Third Law

A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

The third law introduces self-preservation as a guiding principle.

Robots are programmed to prioritize their own survival, but not at the expense of human safety or disobedience to human commands.

Fill in the blanks

1. _____ is the branch of technology that deals with the design, construction, operation, and use of robots.
2. Robots play a crucial role in making tasks _____ and more efficient.
3. The arm-like structure of a robot, called the _____, consists of joints and links.
4. In the field of healthcare, robots assist surgeons in performing surgeries with greater _____ and precision.
5. Feedback mechanisms allow the robot to adjust its actions based on information received from _____.

True / False

1. Robots are only used in manufacturing and industrial settings.
2. Sensors in robots are used to gather information about their environment.
3. Industrial robots are designed to perform tasks that require creativity and complex decision-making.
4. The term "Robot" was first introduced in the 20th century by Isaac Asimov.
5. Humanoid robots are robots that have human-like intelligence and emotions.

Answer the Following Questions

1. Define robotics and importance of robotics.
2. Briefly explain the history of robotics.
3. What are humanoid? Give an example.
4. Describe any 3 key components of a robot.
5. Name the different type of robots.

SKILL SHEET KEYS

Unit – 3: Understanding Robotics

Fill in the blanks

1. Robotics
2. Automated
3. Manipulator
4. Accuracy
5. Sensors

True or False

1. False
2. True
3. False
4. False
5. False

SKILL SHEET KEYS

Answer the following Questions

1. Define robotics and importance of robotics.

A. Robotics is the branch of technology that deals with the design, construction, operation, and use of robots.

Robotics is important because it helps us design machines that can make our work easier, faster, and safer. Robots can do tasks that are difficult, dangerous, or boring for humans. Learning robotics also helps students improve problem-solving, creativity, and teamwork skills.

2. Briefly explain the history of robotics.

A. The term "Robot" was introduced in 1920 by Czech writer Karel Capek in his play "R.U.R." (Rossum's Universal Robots).

In the 1960s, the first real industrial robot called Unimate was made. It was a strong and accurate robotic arm that helped do repeated factory tasks.

In the 1970s–80s, scientists built special labs called robotics research institutes. In these labs, experts worked to improve robots and taught them how to think and learn.

In the 1990s–2000s, robots became smarter. They used sensors to see and feel their surroundings and could make their own decisions using Artificial Intelligence (AI).

From the 2010s to today, modern robotics includes many types of robots like factory robots, surgical robots, self-driving cars, drones, and helper robots. Robotics is still growing with new ideas like soft robots and better ways for humans and robots to work together.

SKILL SHEET KEYS

3. What are humanoid? Give an example.

A. Humanoids are robots that look and act like humans. They mimic human behavior, and can perform activities, such as running, jumping, carrying objects. Some of them can mimic human expressions as well. Robots have sensors embedded and have cameras that enables them to see clearly.

Example: Sophia is a humanoid which exhibits human like expressions and can interact with people.

4. Describe any 3 components of a robot.

A. Robots consist of various components that work together to perform specific tasks.

Power Source: Robots need a power source to operate. This can be in the form of batteries, electric power, or in some cases, hydraulic or pneumatic systems.

Actuators: Actuators are responsible for the movement of robot parts. Electric motors, hydraulic cylinders, or pneumatic pistons act as actuators, allowing the robot to move its joints or other components.

Sensors: Sensors provide robots with the ability to perceive and gather information about their environment.

SKILL SHEET KEYS

5. Name the different types of robots.

A. There are various types of robots designed for different purposes and applications.

Industrial Robots: These robots are strong and accurate arms that help build things in factories. They can weld, paint, and assemble parts. The robot arm, called a manipulator, has joints and links and holds tools like grippers or welders to do different jobs.

Medical Robots: Robots used in medicine can assist surgeons in surgeries or deliver medications. They help make medical procedures more precise and less invasive.

Educational Robots: They can help you understand and practice different subjects, especially in science and programming.

Entertainment Robots: They help in playing games along with you, or even robots used in theme parks.

Domestic Robots: These robots help with tasks at home. They can vacuum the floors and robotic lawn mowers.

Agricultural Robots: Assist in agricultural tasks such as planting, harvesting, and monitoring crops.



THANK YOU